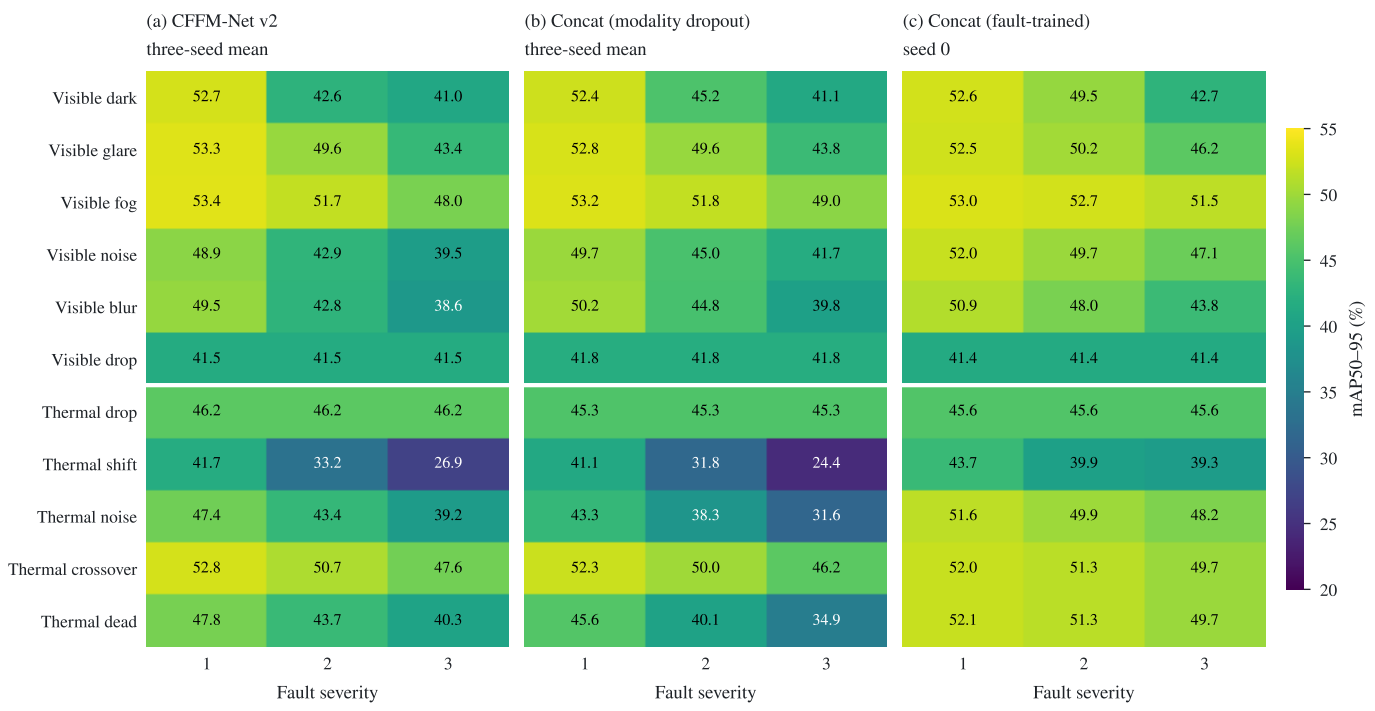
(a) CFFM-Net v2
three-seed mean
(b) Concat (modality dropout)
three-seed mean
(c) Concat (fault-trained)
seed 0
Visible dark
Visible glare
Visible fog
Visible noise
Visible blur
Visible drop
Thermal drop
Thermal shift
Thermal noise
Thermal crossover
Thermal dead
52.7 42.6 41.0
53.3 49.6 43.4
53.4 51.7 48.0
48.9 42.9 39.5
49.5 42.8 38.6
41.5 41.5 41.5
46.2 46.2 46.2
41.7 33.2 26.9
47.4 43.4 39.2
52.8 50.7 47.6
47.8 43.7 40.3
52.4 45.2 41.1
52.8 49.6 43.8
53.2 51.8 49.0
49.7 45.0 41.7
50.2 44.8 39.8
41.8 41.8 41.8
45.3 45.3 45.3
41.1 31.8 24.4
43.3 38.3 31.6
52.3 50.0 46.2
45.6 40.1 34.9
52.6 49.5 42.7
52.5 50.2 46.2
53.0 52.7 51.5
52.0 49.7 47.1
50.9 48.0 43.8
41.4 41.4 41.4
45.6 45.6 45.6
43.7 39.9 39.3
51.6 49.9 48.2
52.0 51.3 49.7
52.1 51.3 49.7
1 2 3
Fault severity
mAP50–95 (%)
55
50
45
40
35
30
25
20